



Review

Aims and methods of biosteganography

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ABSTRACT

Applications of biotechnology to information security are now possible and have potentially far reaching political and technological implications. This change in information security practices, initiated by advancements in molecular biological and biotechnology, warrants reasonable and widespread consideration by biologists, biotechnologists and philosophers. I offer an explication of the landmark contributions, developments and current possibilities of biosteganography—the process of transmitting secure messages via biological mediums. I address, (i) how information can be stored and encoded in biological mediums, (ii) how biological mediums (e.g. DNA, RNA, protein) and storage systems (e.g. cells, biofilms, organisms) influence the nature of information security, and (iii) what constitutes a viable application of such biotechnologies.

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1. Introduction

The new use of DNA encoding technology in information security contexts deserves another coinage. Biosteganography: a field with the aims of traditional steganography—concealing messages—yet with a stark divergence of methods and mediums that have been borrowed and redesigned from genetics, molecular biology, biochemistry and computer science. The available resources for manipulating, transporting and hiding data have changed along with biotechnology. Until recently, DNA encoding—the process of encoding arbitrary data in DNA sequences—has been limited to *in vitro* proof of concept experiments for technological, archival, industrial and computational

applications (Clelland et al., 1999; Goldman et al., 2013; Smith et al., 2003; Palacios et al., 2011; Church et al., 2012; Figureau et al., 2000). This is a conservative assessment of the theoretical and industrial implications of these results—of these industrial implications, for instance, some are undeniably military-industrial. As well as such methods being limited by their intended applications, a too narrow focus on classical, “Central Dogma” molecular biology has stunted the field. Developments from within genomics, metagenomics, synthetic biology and modern biochemistry are fast becoming relevant and deserve consideration within the broader context of biological information security. Moreover, even given the status quo, the technological and political implications of using biological mediums are poorly understood and dangerously under-theorized.

The notion of using microscopic polymers as information carriers is not new. In 1964 Mikhail S. Neiman was the first to propose the use of microminiaturized polymers as digital information storage mediums (Neiman, 1964, 1965). Examples are piling up. In 2012

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George M. Church developed and applied a strategy to encode arbitrary digital information in long DNA segments (Church et al., 2012). In 2013 Goldman and Bertone encoded, stored and decoded digital information from a variety of file types (JPEG, PDF, MP3) in short overlapping DNA segments (Goldman et al., 2013). Clelland et al. (1999) took one of the first leaps in a merger of 1940s style espionage and 21st century biochemistry—DNA microdots: information carrying genetic samples fitting on the period of a sentence. This modern take on a war-time favourite was constructed as a small number of coded DNA sequences concealed within a background of fragmented material from the human genome and spotted on filter paper.¹ The authors were not without an appreciation for an ominous choice of words, “June 6 Invasion: Normandy” was encoded in a 109 residue long oligonucleotide, offering a significant size improvement over papers strips and miniaturized photographs. A span of less than 50 years has seen the realization of a—formerly theoretical—possibility of biological data mediums.

The ways in which information can be extracted from biological entities is changing, rapidly, and in ways essentially unforeseeable by traditional information security groups. Indeed, even commonplace molecular biological methods offer steganographic and cryptographic applications that could potentially occupy a special place in information security and transmission; but molecular biology has progressed significantly beyond commonplace nucleic acid synthesis, sequencing and transgenics. Today we are in a post-genomic, meta-genomic era, we have seen an advanced computational turn in biological modelling and may yet see an overhaul of regulation and production of synthetic genes (Carlson, 2009; Medini et al., 2008).

It is not technically challenging or expensive at present to engineer an *in vivo* storage system for long DNA segments like those of Church et al. (2012). The use of a harmless yet persistent microbe or virus presents itself as an efficient and secure means of transporting encoded information. No X-ray, infra-red scanner, chemical assay or body search will provide any immediate evidence that a large quantity of information is concealed (see Section 5.2). Add to this the complete absence of long distance data interception methods—as exist ubiquitously for electromagnetic devices—and you obtain a virtually undetectable way to carry data. DNA microdots and microbial data vectors are well within the bounds of current technology and are not exceptionally difficult to transport from the lab bench to the military industrial complex.

But the significance of using a biological medium is not just an advancement past 20th century benchmarks of size. Biological information storage and transmission technologies differ significantly from digital mediums due to the possibility of profound amplification of message signal (Figureau et al., 2000; Clelland et al., 1999; Nawy, 2014) and the reorganization of the expertise and infrastructure involved on all sides. The use of codes based on feats of mechanical and electrical engineering can be supplemented, supplanted or bypassed by those based on biochemical and biotechnological ingenuity; code-breaking mathematicians can be bypassed by high-precision sequencing technology and molecular biology. A new steganographic “arms race” may see more valuable contributions from today’s Craig Venters than our Oppenheimers, Einsteins and Turings.

In Section 2, I review the state of the art of DNA encoding technology. I discuss simple codon replacements, more

sophisticated error correcting codes, and storage within protein coding regions. Section 3 covers some biocryptographic methods, cryptographic techniques that could be, or have been, applied *in vivo*. Special attention is given to exemplars of evolutionary, structural, and sequence inspired techniques. In Section 4 I begin to examine steganographic techniques uniquely applicable to biological systems. Possible applications of splice variation, transgenics, and metagenomic techniques are discussed. Section 5 is an assessment of the feasibility of biosteganography, as well as its technological (Section 5.1) and political (Section 5.2) implications.

2. Methods of encoding

Advancements in sequencing technology have occurred rapidly and with profound implications, not the least of which is the quantity and rate of sequencing that can now be done (Chan, 2005; Carlson, 2009; Jain et al., 2015; Loman and Watson, 2015). Complimenting this progress in sequencing—biological information extraction—there has been a parallel leap in the ability to engineer nucleic acids. From the 1959 Nobel prize awarded to Severo Ochoa and Arthur Kornberg “for their discovery of the mechanisms in the biological synthesis of ribonucleic acid and deoxyribonucleic acid” (Nobel Media 2014), to modern synthesis of “functional designer eukaryotic chromosomes” (Annaluru et al., 2014), we have entered a post-genomic-synthesis era, where the precise engineering of large oligonucleotides is no longer a trend-setting abnormality. Moreover, with developments in the CRISPR/Cas9 platform, targeted insertion of designer DNA into vector organisms has never been easier (Hsu et al., 2014; Doudna and Charpentier, 2014).

Producing encoded DNA (enDNA) is firstly a matter of finding a way to unambiguously transform a binary string—the message—into the quaternary code of DNA nucleotides, and synthesizing this resultant DNA strand. Secondly, since a biological medium imposes certain constraints not present *in silico*, some effort must be made to tailor the encoding method to its biological context. Methods diverge rapidly. I will discuss pre-encoding methods, redundancy reducing codes, protein encoding and some methods and problems of increasing the available genomic resources for information storage.

Whatever is meant by “information” in a biological context, i.e. “genetic information”, it is certainly of a different kind than what is meant in digital contexts (Godfrey-Smith and Sterelny, 2008). Nonetheless, nucleic acids like DNA do have properties amenable to digitization of more traditional information: they have sequences in base 4. Consider the simplest case of treating a binary string as a number, then merely converting this number into quaternary and replacing each of 0, 1, 2, 3 with one of A, T, G, C. The problem with this method is not mathematical but biochemical. There is a possibility for long strings of repeated nucleotides to be encoded, thus introducing a greater possibility for synthesis or sequencing error, since most sequencing methods still do not work well on long repetitive regions (Goldman et al., 2013; Church et al., 2012).

Although there have been profound advances in both sequencing and synthesizing long DNA segments, synthesis lags behind sequencing. So we should first be aware that there are methods that do not require oligonucleotide synthesis. The use of a book cypher—or genome cypher in this case—accomplishes just that. A book cypher uses a book as a key; the encoded message is typically sent as a string of numerical triplets where the first number specifies the page, the second the line and the third the word on that line. Similarly, one could send an electronic message as a text specifying the set of genomic loci in which to look for an encoded sequence. The large genomes of some eukaryote could easily be used, perhaps by specifying a triplet of chromosome, reading frame and a range of nucleotides to include.

¹ There are commercially available applications of similar technologies for ‘theft prevention and recovery’ available from SelectDNA™, among others. The intended application involving a spray or capsule (see patent of Knights et al., 2013) filled with synthetic DNA that is projected at a suspected thief to allow later identification by sequencing samples taken from clothing or skin. Although it should be noted that any sufficiently specific DNA sample would suffice for the purpose described.

Indeed, the use of a genome as a one time pad is similar to a genome cypher in spirit (See Section 3, [Gehani et al., 2004](#); [Clelland et al., 1999](#)). The choice of the genome-key becomes the weakest link. One could always choose the genome of an organism already located in some public database, a newly discovered and not yet public organism, or, to avoid the need to discover new organisms, one could simply mutate an existing organism to alter the reading frame outside of coding regions. Choosing an organism whose genome is private does necessitate sending the organism and the cypher text separately and performing whole genome sequencing, yet in many contexts this will be a small price to pay for the added steganographic security of the biological medium (see Section 4). Indeed, these quasi-digital methods still suffer from many of the same the security threats characteristic of digital transmission.

[Goldman et al. \(2013\)](#) employed a DNA encoding method using a Huffman code—a method that minimizes sequence redundancy ([Huffman, 1952](#))—to encode arbitrary digital files in short overlapping DNA molecules. In addition, the enDNA oligos were capped on each end with sequences that facilitated parity checking, thus allowing subsequent verification of data integrity upon sequencing ([Goldman et al., 2013](#)). This method is much less likely to produce problematically repetitive sequences—information extraction is not error prone—which is precisely the kind of encoding desired for a data storage system. The Huffman code also has the advantage that it encodes data in minimal-sized sequences. This DNA encoding method and two others were originally discussed in [Smith et al. \(2003\)](#). One of these alternative methods involved the punctuation of base sequences by a regularly repeating nucleotide. As the authors note, this method produces a sequence that is obviously artificial: an advantage in databasing contexts and disadvantage from an information security perspective. [Smith et al. \(2003\)](#) also described a method that made use of alternating purines and pyrimidines, so that the final sequence always had an equal number of each—a property that makes amplifying DNA by polymerase chain reaction (PCR) easier.

Exploiting the degeneracy of the amino acid code has led to a more biologically inspired encoding method. [Arita and Ohashi \(2004\)](#) published the first “biological innocuous” method of encoding data within protein coding DNA (pcDNA)—leaving the resulting protein structure the same while changing the underlying nucleotide sequence. This method, they say, was developed to “help establish brand names for the engineered strains and to resolve legal disputes regarding gene-related patents”, although its application to information security is clear: one can encode information in DNA without apparent modifications to even so much as the proteome. [Haughton and Balado \(2013\)](#) offer another method of encoding data in pcDNA that, unlike that of [Arita and Ohashi \(2004\)](#), does not require the recipient have knowledge of the original nucleotide sequence. Both of these methods, like those of [Heider and Barnekow \(2007, 2008\)](#) and [Liss et al. \(2012\)](#), were developed to aid in the process of DNA watermarking *within* transgenic protein sequences. As well as adding to the available genomic encoding resources, this technique may help circumvent some side effects creating transgenic vectors discussed below.

A message need not be encoded at a single genomic locus. To avoid the need for an error correcting code, one could copy the same message into multiple places throughout the genome and then extract this message by whole genome sequencing and alignment ([Yachie et al., 2007](#)). This method of encoding is another that is more suited to long term DNA data storage, since the redundancy of DNA message encoding ensures message stability even with a high mutation rate, yet also increases the chance of discovery.

Location is key. Before the pcDNA encoding techniques of [Arita and Ohashi \(2004\)](#) and [Haughton and Balado \(2013\)](#), it was thought that the vast resources of junk DNA offered a means of storing data in phenotypically indistinguishable organisms. Nonetheless,

the functional significance of junk DNA is still debated today ([Graur et al., 2013](#); [Eddy, 2012](#); [Kellis et al., 2014](#)), and not all sides agree on the expected consequences of modifying, removing, or introducing foreign DNA—even in the extremely bloated genomes of some eukaryotes. Consider the worry of [Morford \(2011\)](#) on the potential side effects of introducing enDNA into active cells.

Message may be translated, resulting in potential “superbugs.”

Also, an intruder may purposefully examine the bacterial colony for cells that have morphological and other anomalies that result from the side effects of encrypting a DNA message in a living organism.

While the threat of biosteganographic “superbugs” is likely quite low, unless the vector is closely related to a pathogen, there is certainly a possibility that unintended translation or interference with nearby regulatory regions could have other anomalous effects: not the least of which is lethality or inhibition of the intended vector organism. pcDNA encoding techniques can help to circumvent this, but additional constraints are required. Although the genetic code is redundant—multiple codons specify the same amino acid—individual organisms have biased usage of specific codons, i.e. preferred codon ratios that introduced genes must satisfy to avoid anomalous depletion of pools of cytoplasmic tRNA (See [Romanos et al., 1992](#); [Kudla et al., 2009](#)). While [Liss et al. \(2012\)](#) seem to have been the first to attempt to account for this constraint on codon ratios, their method will not preserve codon ratios when variation in preferred codon ratios is high for a given amino acid—a problem case accounted for by [Haughton and Balado \(2013\)](#).

Even without sophisticated compression and pcDNA encoding techniques, the quantity of data that can be stored per unit volume in biological mediums far exceeds the capacity of magnetic tape and disks. Since there are 4 nucleotides, each nucleotide can store 2 bits as a binary string, ex/ A = 00, T = 01, C = 10, G = 11. So a set of 4 nucleotides can store 1 byte, and 4 Mbp is thus about 1 Mb of digital data—even with this naïve encoding scheme, i.e. without compression or any of the encoding advancements described in Section 2. Since a yeast artificial chromosome (YAC) inserted into *S. cerevisiae* can be about 1–2 Mbp ([Burke et al., 1987](#)), if we assume that we can replace the majority of these bases (excluding perhaps the origin of replication) with those of our choosing, then we could store about 500 kb of digital information or about 500,000 characters. If we take the upper bound for a novel or dissertation to be 100,000 words, since the average word length in English is about 5 characters per word then a YAC can store a novel. Of course, this is an underestimate since various compression techniques could significantly increase this number. [Cox \(2001\)](#) comes to a similar conclusion, but adds that if we could store 10^8 YACs in some kind of databasing capsule then we would have far exceeded the average library with its mere 10^5 books. Indeed we could easily fit that many yeasts in a microscopic capsule, but subsequently extracting this information would be problematic unless there were many duplicate copies of each strain or single cell sequencing methods were to drastically improve ([Nawy, 2014](#)). To my mind, the number of strains we collect together is arbitrary and 500,000 uncompressed characters in a microscopic vector is more than appropriate for a secure hidden message.

Looking forward, there is potential to also encode information over a DNA sequence using patterns of cytosine and adenine methylation. Such a method may indeed evolve out of synthetic epigenetics, defined by [Jurkowski et al. \(2015\)](#) as, “the design and construction of novel specific artificial epigenetic pathways or the redesign of existing natural biological systems, in order to intentionally change epigenetic information of the cell at desired loci.” The use of epigenetic encoding ([de Groote et al., 2012](#); [Jurkowski et al., 2015](#)) to supplement or supplant DNA encoding is a more remote possibility at present than even the most elaborate forms of DNA based

steganography. Indeed, as [Jurkowski et al. \(2015\)](#) put it, “Bottom-up synthetic epigenetics...[is] still in its infancy.” Nonetheless, the possibility of increasing the number of bits of information per nucleotide stored (e.g. in DNA databases) cannot be overlooked.

While junk DNA may provide some (marginal or significant) resource for DNA encoding, artificial plasmids or chromosomes certainly extend DNA encoding resources ([Annaluru et al., 2014](#); [Cox, 2001](#); [Burke et al., 1987](#)), and the possibility of total synthesis of artificial genomes ([Gibson et al., 2008](#)) seems a sort of limiting point on the *in vivo* storage of enDNA. Furthermore, pcDNA and epigenetic encoding techniques applied in conjunction with expansions of available encoding regions could only increase or supplement existing space resources. It is clear that there has been significant advancement past simple methods of numerical transformation from binary to quaternary, all of which are afflicted with biologically prohibitive lengths or repetitions. Redundancy reducing methods overcome these problems, but at the cost of sequences that do not appear biological in origin: sequences that risk being easier to identify by interceptors. Finally pcDNA encoding strategies—in conjunction with genomic resource expansions—offer a less biologically disruptive means of encoding within the genomes of living organisms.

3. Biocryptography

If someone can encode data in DNA, then of course many methods from traditional cryptography will be transferable without remainder. Nonetheless the biological medium presents its own challenges and advantages. With exploitation of the specifically biological aspects of DNA encoding we step firmly into the realm of biocryptography. There is a growing field of *biologically inspired* cryptographic methods that are beyond the scope of this paper (see [Ibrahim and Maarof, 2005](#); [Gao, 2011](#); [Hordijk, 2005](#); [Ramana et al., 2015](#)). Indeed, while it is true many such methods could be implemented *in silico*, their application *in vivo* provides a unique opportunity, among much else, to restrict those capable of interception to a group with bio-molecular expertise (see Section 5.2). The inspiration for these methods fall into evolutionary, structural and sequence based biological processes.

Consider a context where the recipient of the DNA message is sure to be able to sample the DNA correctly, but an interceptor might fail to sample correctly and obtain only a fraction of the DNA intended for transmission (as might obtain if fragments of the message are located on different parts of the same object). In such a case it is advantageous to ensure that each part of the message cannot be decrypted in isolation. Familiar to molecular phylogenetics is the notion of reconstructing an ancestral sequence from extant sequences given an evolutionary model ([Thornton, 2004](#))—with some knowledge of how the sequences evolved from their ancestor, one can reasonably infer the ancestral sequence. Run in reverse, one can apply an evolutionary model to a given sequence and obtain a set of “evolved” sequences divergent from the original. One could then transmit these divergent sequences to a recipient with knowledge of both the model used to mutate them and of the proper way to sample a sufficient quantity of them—a two key system. Here the evolutionary model of the mutational process is serving as a probabilistic key: a key sufficient within a margin of error to reconstruct the message sequence given enough of its descendants.

The potential use of a model of a biological process as a key is certainly not unique to probabilistic or evolutionary models. The burgeoning field of protein structural prediction has much to offer in this respect, especially *because* protein structural prediction is still an open problem ([Moult et al., 2014](#)). Presuming for the moment that one has a way to encode arbitrary data in the 3-D structure of a folded protein, a message could be passed as the

DNA segment that codes for that protein given a particular model of folding. Say the data was encoded as a string specified by the order of connections between amino acids running from the N- to C-terminal ends of the protein (a “contact matrix”). Granting an interceptor knows the data is to be extracted from the folded protein and not the sequence, even slight changes in choices for the parameters² of the protein folding model could drastically alter the order of connections along the amino acid backbone and thus the encoded message. Of course, designing proteins with highly specific folds *in vivo* is beyond the status quo of biochemistry, but designing models of protein folding that give specific and predictable results is not beyond the reach of bioinformatics.

What is currently out of reach in the case of protein chemistry may not be for long; consider the case of nucleic acids being used to the same end. The use of structural features of biological macro-molecules has been taken up by [Halvorsen and Wong \(2012\)](#). The authors demonstrate a method of encoding messages by exploiting the “geometric conformation of DNA nanostructures”, wherein each bit of information is transmitted as an oligonucleotide of a given length in one of two conformational states—states that are assumed only once an appropriate molecular catalyst has been introduced ([Halvorsen and Wong, 2012](#)). The authors also note some of the advantages and disadvantages of using biological mediums, such as the requirement for “physical message...technical skill, laboratory equipment, and time” (p.2) addressed later (see Section 5). Indeed, we are now in a position to see the laboratory implementation of formerly theoretical biological processes.

[Figureau et al. \(2000\)](#) describe two methods that they call, “a physico-chemical realization of a secret key cryptographic system.” The first method involves Alice sending an encoded message in a long DNA molecule to Bob, where the message is flanked by a promoter made from Bob’s DNA and a known restriction enzyme cut site. When Bob receives the message he cuts the DNA with the appropriate restriction enzyme, then selects from the mixture of DNA and sequences only those fragments that can hybridize to his DNA. An interceptor (Eve) will have a difficult time determining where to sequence unless she knows both the flanking restriction enzyme cut site (EcoRI for example), and sequences of Bob’s DNA—indeed, the problem for Eve could easily become intractable if the message is small enough, the DNA is long enough, and peppered with segments of human DNA³ (see [Leier et al., 2000](#) for a precise calculation). The second method suggested by [Figureau et al. \(2000\)](#) involves Bob pre-emptively sending a sample mixed with meaningless DNA, containing restriction enzyme cut sites between two promoters crafted from his own DNA. Bob also radiates the DNA to form pyrimidine dimers: a fusion of adjacent thymine or cytosine nucleotides that can impair sequencing. When Alice receives the sample she chemically treats the sample to remove the pyrimidine dimers, inserts her message between Bob’s promoters by cutting it with the appropriate restriction enzyme and ligating the sample back together, then sends the sample back to Bob. An interceptor will be unable to amplify the correct region without knowing the promoter sent by Bob—something that even Alice does not know. The idea of using primers as a key to amplify a DNA segment by PCR has been taken up by a number of authors (see [Leier et al., 2000](#); [Anam et al., 2010](#)), and can be combined with the

² The parameters for this model need not be chosen to reflect any sort of biological meaningful context for protein folding, and in fact the cypher is more secure the less these parameters are predictable. All that are required from the parameters is that they fold the protein into something with a sufficiently rich structure to allow data to be encoded therein.

³ A random primer could be used, but attempting to design something effective in a heterogeneous mixture of DNA from multiple species is unlikely to amplify anything but noise.

use of a selectable marker—a transgene that allows selection of one bacterial colony from a mixture—to create a two-key cryptosystem (Morford, 2011).

While biocryptography has been widely addressed elsewhere, it serves to note some features of modern biological science that can be applied in conjunction with biosteganography. Models from molecular phylogenetics can be used to produce evolutionary-keys, allowing the separation of a single message into multiple, individually unreadable, divergent messages. Bioinformatic protein structural modelling can play a role analogous to security by obscurity: hiding messages behind difficultly discerned folding parameters. And advancements of sequence manipulation and amplification can facilitate the transmission of a message in complex mixtures where *a priori* experimental extraction would be intractable.

4. Biosteganography

The following expands on those aspects of biological information security that are *specifically* steganographic. On the boundary between nucleic acid biochemistry and steganography are opportunities to exploit *in vivo* biological processes that manipulate sequence information in repeatable ways. Consider the alternative splicing of mRNA transcripts—a eukaryote specific mechanism whereby introns in mRNA are cut out and the remaining exons are spliced together to produce a new sequence. If splicing were identical across all organisms it would not add much to our information security, since an interceptor could just insert the extracted message into any old eukaryote, allow splicing to occur, then obtain the message (or just simulate the process). But splicing occurs differently in different species, sexes of the same species, and cell types of a single organism. For example, the *dsx* gene in *D. melanogaster* has exons 1, 2, 3, 5, and 6 spliced together in males, and 1, 2, 3, and 4 in females (Lynch and Maniatis, 1996). So designing a message to be extracted as a post-splicing mRNA in male *D. melanogaster*, would give an incomplete message if extracted as transcriptomic data in females. Looking forward, one can imagine combining epigenetic encoding (to influence cell type) and cell type specific splicing in a manner akin to the two-key crypto-system of Morford (2011).

Of course, if one knows the potential splice patterns it would be easy to simulate splicing *in silico*, but this is not mathematically or biological trivial. Indeed one might think of choosing the appropriate organismal background to decoding as a biological equivalent to computational hardness. Only once an interceptor has established a set of specifically biological characteristics of the data can one begin to attempt to decipher the message—a message buried in sequence information that may have already been traditionally encrypted and difficult to obtain. This is as if a piece of information was on one of several computers (organisms), each computer had multiple hard-drives (chromosomes, plasmids, etc.), and each hard-drive could be read in six different directions (reading frames), but the message could only be decoded using the proprietary code on another system. The message might be stored in one eukaryote but need to be expressed (interpreted) in another, such as being transported in *C. trachomatis* but interpreted in female *D. melanogaster*. That methods of this type are steganographic derives from the difficulty of even determining the location of the message from amongst a variety of potential splice variants.

We have primarily been considering the transmission of data in biological mediums as the insertion of a DNA segment into a vector and subsequent transport on an object of some kind. This base case is quite limiting when compared to the complex ways in which biological entities propagate themselves, and the ways biological data is gathered today.

Transporting data in biological mediums can utilize host-vector and host-host interactions to expand transmission routes—something we might term “epidemiological steganography.” While host-vector interactions might of course be antagonistic, there are plenty of transient, persistent, and benign host-vector relationships that could facilitate temporary host⁴ based transport. The exploitation of an unsuspecting intermediary and recovery from a non-targeted actor has parallels in illicit transport of all kinds, but the use of a microorganism or virus⁵ adds the capacity for the intermediary to unconsciously, intermittently and transiently create others. A message could easily be propagated along a lineage of host–host vector exchanges.

Gathering data by metagenomic shot-gun sequencing—sequencing small segments from multiple genomes at the same time then reconstructing individual genomes afterwards—provides another way to store parts of a message in multiple places, each being insufficient to reconstruct the whole (Section 3). Consider the possibility of storing or transferring information as a set of partially overlapping sequences within a multiplicity of species or a biome. Sequencing any particular organism would provide only a useless fragment of a larger message, but doing a complete shot-gun sequence would recover data as a “ghost organism”: sequence reads from multiple organisms that cluster together due to high overlap. Depending on the parameters of clustering and the experimental set up for data recovery, a technique like this holds promise of allowing data transmission in such a highly disperse form as to be essentially unrecognisable as data.

To my mind, the line between cryptographic and steganographic uses of biological mediums in information security is thin. This is because the use of a microbiological or macromolecular medium to carry data is practically steganographic by itself: microbes and nucleic acids are small and easily transferred without notice. Nonetheless, I have discussed some ways to exploit biological mediums to hide data that employ more than their size and obscurity. In the remaining section I will address those aspects of the biological medium that lend themselves to application in security contexts.

5. Biological mediums

Henry Fountain wrote in the *New York Times* of DNA microdots (Clelland et al., 1999) that, “The technique seems less likely to be used by a real-life James Bond, unless he has a degree in molecular biology.” Indeed, excluding the fact that plenty of people have such education, the use of a biological medium does not lend itself well to unrealistically paced fantasies about high-tech espionage. Although that surely only indicates that this is hardly the proper mode of comparison. Moreover, that traditionally trained military personnel would be unable to use or misuse data in biological mediums can be quite an advantage (see Section 5.1).

Modern technology driven espionage is less about direct conflict and its associated gadgetry and more about persistent infiltration and maintaining secure lines of communication—an area in which biological mediums indeed have something to offer. One need not take anyone's word in this matter, merely look at the research funding practices of major intelligence and military industrial organizations. DARPA, for example, recently funded a project by Palacios et al. (2011) to, “write and encode data using arrays of genetically engineered strains of *Escherichia coli* with fluorescent proteins (FPs)

⁴ A candidate host for a biosteganographic vector could easily be a plant, animal, or any organism with a microbiome to which vectors could be added.

⁵ Note that since a virus has virtually no junk DNA this kind of transmission would be impossible were it not for the pcDNA encoding techniques of Arita and Ohashi (2004) and Haughton and Balado (2013).

as phenotypic markers”. Hardly a flashy spy-gadget, but surely an indication of the extent of biological research spending by intelligence and military industrial groups.

Prognostications aside, a proper niche is required for the application of biosteganographic techniques. There are five potential contexts in which the use of biosteganographic techniques could become more than an academic hypothetical, i.e. contexts that would encourage the development of the technical requirements for biosteganography. (i) If increased need for the diversification of security practices emerged, (ii) if solutions were found for canonical hard-problems in digital cryptography, (iii) if significant developments occur in methods of interception at the site of encryption or decryption (“side attacks”), thus overcoming the need for computational solutions, (iv) if digitally or mechanically prohibitive contexts were encountered, such as digitally monitored transmission routes and (v) if sufficient funding were directed by interested parties at either the investigative sciences involved, or the agencies who might employ them, i.e. if motivation was generated financially. Indeed, all of these five contexts are beginning to emerge today to some extent.

Biocryptographic and biosteganographic techniques will surely be developed beyond their current state, if for no other reason than that they are parasitic upon the development of techniques required for other aims and methods. Moreover, the requisite molecular techniques are becoming not only more advanced but also more cost effective every year (see Carlson, 2009). Put another way, the feasibility of biosteganographic and biocryptographic techniques is certainly just that of the biomolecular, microbiological, and chemical industries underpinning their implementation. What remains uncertain at this point are the technological and political implications of these industries.

5.1. Technological implications

The interplay between longevity/stability and time-sensitivity/instability of the biological medium determines its utility in either databasing or steganography. Highly stable DNA storage methods are more applicable to implementations in databasing, while ephemeral enDNA transfer protocols are more suited to biosteganography. Indeed, this means that the time sensitivity of biological information can now play a role that is not available to digital or solid state transmission mechanisms. DNA as a data storage medium is quite durable: it can persist in an appropriate solution for millennia (Goldman et al., 2013; Cox, 2001; see Grass et al., 2015 for calculations of enDNA longevity when stored in silica). Indeed, as Cox (2001) says, enDNA as a data storage medium, “has a proven track record (life)” and is “unlikely ever to become obsolete.” But the stability of DNA is a double edged sword: enDNA can potentially serve as a reservoir for transmitted information that is difficult to completely erase without proper chemical or biological agents. The utility of stable DNA is a point of divergence for those interested in enDNA as a databasing material, and as a secure data transmission material. The use of spore forming vectors like *B. subtilis* and *S. cerevisiae* or error correcting methods of encoding only add to this lifetime (for a discussion of the possibility of error correcting codes in living organisms see Liebovitch et al., 1996).

For steganographic purposes, transmission of an enDNA molecule within a living organism with an error-prone code has some advantages. Firstly there is no requirement to employ any standard storage vessel that might arouse suspicion; one need only appropriately choose the biological vector to be hardy enough to survive transport. Nonetheless a vector that can survive transport raises the worry that it might survive long after transport—that is, data that can literally run off on its own (see Wright et al., 2013

for transgene security measures given existing technologies). Fortunately, the active division of vector cells ensures that mutations will eventually be introduced, compromising efficient decoding. While wild type vectors are not guaranteed to degrade on a tactically significant time-scale, genetically engineered strains could conceivably shorten or specify this period. Looking forward, our growing knowledge of programmed cell death (Fuchs and Steller, 2015) or synthetic epigenetics (Jurkowski et al., 2015) has potential to lead to genetically engineered self-destructing vectors, and the potential for alternate (xeno) amino acids⁶ could at least prevent horizontal transmission to environmental organisms (for an opposing position see de Lorenzo, 2010).

Transmission of a message as a sample of more fragile RNA also avoids the problem of the superfluous stability of *in vivo* DNA. RNA degrading enzymes, ribonucleases, are ubiquitous in nature and could be relied on to dispose of a message to avoid security breach. However one exploits the activity of biological mediums to achieve time sensitivity, this is surely an avenue not open to the transport of digital information unless it is actively being read and overwritten—hardware processes that are arrested by conventional means.

As well as accounting for vectors that self-destruct, contingencies need to be developed for the possibility of encountering a data vector chosen to be hazardous, or even lethal, if not properly handled during interception. Consider, for instance, transporting a message encoded into a toxic or irritating microorganism. A careful recipient could isolate the vector's genome and extract the message with relative ease, but a careless interceptor might face unexpected medical complications. Indeed, frightful situations like these provide further impetus to develop the capacity to assess the microbiological and pharmacological capabilities of potential senders, receivers and interceptors.

There is currently no way to scan an object or organism from a distance with the resolution required to detect vector microorganisms. Put another way, microorganisms have no detectable emanations sufficient to indicate the presence of encrypted data. This may seem an obvious point to biologists, but the fact that digital systems like hard-drives and smart-cards have detectable emanations—i.e. are subject to side attacks—remains a serious security threat (Tanaka, 2008). Information leakage in the form of optical, acoustic and electromagnetic radiation is addressed in the military Tempest project (although possibly known to the Soviets as far back as 1954) (Boak, NSA, 1973), and completely absent in the transmission of data through biological mediums. So biological mediums have the advantage of being secure at distances that digital mediums are not.

Consider a similarly optimistic view expressed by Halvorsen and Wong (2012),

Unlike encryption schemes that rely on mathematical algorithms...biochemical based encryption is not directly vulnerable to increasing computational power.

Although computation may be of no use when extracting information directly from biological mediums, side attacks on biocrypts are still a security threat at the site of encoding and decoding to the extent that computation is required by on-site technology. Nonetheless the nature of side attacks would need to change significantly: methods specialized for use against personal computers and cellphones would need to be reworked for detection of emanations from DNA sequencers, synthesizers and a variety of other specifically biological technologies. Moreover, with some of the methods

⁶ Schmidt and Giersch (2011) also offer up the possibility of xeno amino acids, characterized in Herdewijn and Marliere (2009) and being used to construct artificial, unregulated, xenobiological proteins for industrial applications.

above it would not be impossible to encode, transmit, extract and decode the message with pre-digital⁷ biological methods.

Some technological developments are required to motivate the use of biosteganographic over or alongside extant digital methods. Foremost is the potential for small DNA synthesizers to complement the already miniaturized sequencers: nanopore systems are a great way to receive data (Loman and Watson, 2015), but unless a desktop synthesizer is made available the burden of information transmission and defence from side attacks will rest unfairly on the sender. Secondly, if a biological vector other than a simple oligonucleotide stabilizer is desired as a means of transmission, some advancement (miniaturization, commercial availability) in methods of inserting vector DNA into hosts would be beneficial. For example, an emanation-shielded desktop version of an automated CRISPR/Cas9 platform would make vectorization as easy as printing documents, more secure than microdots and may even facilitate the vectorization of epigenetically encoded information. As Jurkowski et al. (2015) write,

Cas9 systems fused to a selection of different epigenetic modifiers could be simultaneously used in a single experiment to target various epigenetic modifications to selected loci. (p.4)

Finally, and more generally, any decentralization of the industrial synthesis or analysis of biological mediums would remove the risk of third-party security compromises (see comments below on Carlson, 2009).

5.2. Political implications

Although the current cost of synthesis for large DNA segments can be prohibitive, DNA segments capable of encoding large books are certainly not approaching the current theoretical maximum. For instance, Gibson et al. (2008), working at the J. Craig Venter Institute, recently published the synthesis of the complete 1.08 Mbp genome of a genetically modified *M. mycoides*, and Goldman et al. (2013) successfully encoded and decoded 757,051 bytes of data in overlapping 117 nt long oligonucleotides totalling about 18 Mbp.

Although the DNA sequencing industry currently is not as lucrative as other biotechnology industries, we should consider the potential for economic and technological factors to change the nature of sequencing. As Carlson (2009) writes,

The question is whether customers for DNA of a specific sequence will continue to order it from centralized facilities, or whether economic, technical and regulatory factors might contribute to a decentralization of synthesis. New technologies could enable desktop instruments that provide rapid and secure gene synthesis. Similar technological transitions have resulted in profound transformations of the infrastructure we use for computing, printing and communicating, all of which can now fit in a pocket. (p.1093)

We may need to wait for hand-held synthesizers, but the development of nanopore sequencers has already reached such a stage (Jain et al., 2015). As Loman and Watson (2015) report,

Sequencers have become smaller and cheaper and are entering the hands of individual academic labs and clinical environments. The MinION [nanopore sequencer] takes this trend one step further: it fits in your pocket. (p.304)

⁷ Further down the line, if DNA computing reaches a certain level of complexity it may be the case that biocyphers could be constructed within DNA computers, effectively skipping the step of using electronic computers prior to DNA synthesis.

Although since MinION requires sending data to a third party for interpretation, some in-house data analysis would be required before such a nanopore sequencer could be used to decrypt secure information.

Even small changes in our synthesis capacity could bring about profound changes in the industry. Again, consider the perspective of Carlson (2009) on the potential backfiring of regulatory demands placed on centralized sequencing facilities,

Given a choice or if forced by regulatory action to make a choice some designers of new DNA circuits will inevitably conduct business with synthesis providers who do not maintain an archive of design files. (p.1093)

Put another way, we could see the drive of research enterprises to avoid costly regulatory practices instigate a *de facto* black market of DNA synthesis—opening the door to all the potential pitfalls of decentralized unregulated industries with military industrial applications.

It is not only technology that is unregulated and decentralized. Those who currently hold expertise in biosteganography are a non-political group, holding distinct and unrelated positions in various jurisdictions and geopolitical climates. They are a non-group, embedded within academia or private biotechnology firms, and falling under no general classification of allegiance. This is something of a recruit problem. So the application of biosteganographic methods of encoding is limited by the decentralized distribution of expertise and resources. So is the application of counter-biosteganographic methods, i.e. the current absence of a well developed bio-science infrastructure within military-industrial groups indicates an unreadiness of those groups to intercept and decrypt biosteganographic data.

Not only interception of biosteganographic messages requires a reorientation of expertise. Locating leaks, loose links and evaluating facilities in the chain of transmission requires a new forensics when a biological medium is employed. Evaluation of the trustworthiness of those involved (with key, implementation, facilities, etc.) will also depend on the medium. Personnel that could not be trusted around digital data storage centres might pose no threat around DNA based databases—since traditional side attacks are impossible and very specific technologies and methods are required to obtain a sample and extract data. In this respect, the wet laboratory context differs significantly from the computer lab or radio outpost.

Once the requisite expertise is acquired, even relatively financially limited labs could construct currently undetectable biocyphers. For groups under constant electronic surveillance this could mean a partial levelling of the playing field of data security. Even only a partial distribution of resources between traditional and biosteganographic applications would facilitate a diversification of information security practices, potentially lessening the burden of leaks, captures or interceptions. For instance, if the key to digitally transmitted information were occasionally sent along biosteganographic transmission routes—and vice versa—an interceptor would need a diverse set of interception and decryption technologies, practices and expertise.

Biotechnological experimentation is hardly restricted to nations with the military industrial spending of the west. The existence of biochemically advanced facilities for weapons or drug production in a myriad of nations provides ample opportunity for co-optation of current labourers, researchers and technologies to bio-cryptographic and steganographic ends. That biosteganographic expertise is limited by decentralization certainly does not imply that it is in limited supply. Consider similar worries about the capacity to produce biological weapons, expressed by the Canadian Security Intelligence Service (CSIS, 2000).

Weapons of mass destruction, biological agents are easier and cheaper to produce than either nuclear materials or [chemical weapons] agents, and the necessary technology and know-how is widely available. Any nation with a modestly sophisticated pharmaceutical industry is capable of producing [biological weapons] agents.

The development and utility of biological data mediums parallels that of biological weaponry in more than their mutual ubiquity: both are military-industrial consequences of techniques developed within molecular biology.

6. Conclusions

Data can be encoded, encrypted and hidden in biological mediums. DNA encoding has progressed well beyond simple string conversion techniques to the use of compression and redundancy minimization, more biologically innocuous methods (i.e. junk DNA, artificial chromosomes, pcDNA and epigenetic encoding methods), and methods accounting for specifically biological limitations (i.e. concentration of tRNA pools). Encryption techniques from traditional cryptography have been blended with insights specific to structural, sequence-based and evolutionary biology (Section 3). And—aside from the sense in which all data stored in *microorganisms* is steganographic—splicing, obscuration within complex DNA mixtures, host-vector and host-host interactions facilitate the further steganographic use of biological mediums (Section 4). The advancement of the field past the status quo will indeed require advancements in sequencing (i.e. in-house nanopore data processing) as well as synthesis (i.e. miniaturization, decentralization and epigenetic encoding), but the work of many authors has shows us that even current techniques are sufficient to produce effective biosteganographic and biocryptographic systems.

If the biosteganographic arms race does not await us, something like it may await our more biochemically fluent posterity. Given the state of the art, we must not wait for the technology to be in use before we devise the kind of theoretical and technological apparatuses needed for detection and defence, or to prepare ourselves pre-emptively. It cannot be understated that the introduction of new technological capacities—especially dangerous ones—mandates a serious consideration of the potential unintended consequences.

A final word on the introduction of new mediums. New technologies have the capacity to resurrect formerly outdated ones. Nietzsche said, “Thoughts that come on dove’s feet guide the world” (Nietzsche, 1966). In his time, the dove’s foot was indeed the most secure method of communication. We can now envision a future in which something like Nietzsche’s statement holds true, but in which the biological entities enabling communication are microscopic and much more secure. We may see the return to grace of something like the homing pigeon, which would not carry messages on paper strips attached to its feet, but would instead carry them in the gut bacteria of mites helplessly along for the ride. In general, the prospect of biosteganography might encourage new explorations of vintage methods and techniques, a familiar outcome that seemingly occurs after every technological development in media. We have smaller couriers today, spies fitting on the head of a pin or swab of a sore throat.

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